

BOUNDARY RIGIDITY AND STABLE LEFT–RIGHT MODULI AT GENERIC DEGREE THREE

NATHANIEL MONSON

ABSTRACT. We exhibit a one-parameter family of polynomial Keller maps $\mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$, all of generic degree three, whose members remain pairwise inequivalent after adjoining arbitrarily many identity variables and applying arbitrary polynomial changes of source and target coordinates. The family is obtained from the quadratic cubic-frame data

$$A_{\alpha}(c) = c + \alpha c^2, \quad B_{\alpha,\beta}(c) = -2 - 4\alpha c + \beta c^2.$$

For $\alpha \neq 0$, the complete stable left–right invariant is

$$q = \beta/\alpha^2.$$

The line $\alpha = 0$ is a single ordinary left–right orbit, stably distinct from every $\alpha \neq 0$ orbit.

The invariant is recovered from the nonproperness divisor. After normalizing its singular discriminant component, one obtains an affine plane with two intrinsically marked curves. A unique-factorization argument shows that every automorphism of every affine cylinder preserving this marked pair fixes q . A separate singular-incidence argument treats the exceptional value $q = -2$. Exact symbolic calculations verify the discriminant, normalization, conductor, exceptional fiber, and the explicit gauge transformation on the line $\alpha = 0$.

More generally, on the squarefree coprime cubic-frame locus, the complete stable invariant is the deleted infinity scheme $V(A/c)$ decorated by the transverse derivative $B|_{V(A/c)}$. Stable and ordinary left–right equivalence coincide there. This yields stable-moduli spaces of arbitrarily large dimension, all at generic degree three and, for each family, fixed ordinary degree.

Working manuscript. The exact symbolic verifier has been rerun from the recovered source artifact. Independent implementation and specialist review remain submission requirements.

1. INTRODUCTION

The recent counterexample to the Jacobian conjecture gives a polynomial map $\mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$ with nonzero constant Jacobian determinant that is not an automorphism [1, 6]. We credit Akhil Mathew as the primary human source of the problem: Mathew prompted Levent Alpöge, Alpöge prompted Fable, Fable produced the example, and Alpöge announced it publicly [7]. The construction was quickly placed in larger families. In particular, public

Date: July 22, 2026.

accounts allow arbitrary one-variable deformation data and produce counterexamples of every generic degree [7, 3]. Varying generic degree already gives inequivalent maps.

This paper asks a narrower question:

Can a family of counterexamples of one fixed generic degree carry a modulus that survives arbitrary polynomial source and target changes, even after stabilization?

The answer is yes already at generic degree three.

Let

$$F, H: \mathbb{A}^n \longrightarrow \mathbb{A}^n$$

be polynomial maps. They are *polynomially left–right equivalent* if there are polynomial automorphisms Φ, Ψ of $\mathbb{A}^n$ such that

$$H \circ \Phi = \Psi \circ F.$$

They are *stably polynomially left–right equivalent* if, for some $m \geq 0$, the maps

$$F \times \text{id}_{\mathbb{A}^m}, \quad H \times \text{id}_{\mathbb{A}^m}$$

are polynomially left–right equivalent. We use this definition throughout; “stable equivalence” has other meanings in the Jacobian-conjecture literature.

For $\alpha, \beta \in \mathbb{C}$, consider

$$A_\alpha(C) = C + \alpha C^2, \quad B_{\alpha, \beta}(C) = -2 - 4\alpha C + \beta C^2. \quad (1.1)$$

These polynomials define a Keller map $G_{\alpha, \beta}$ by the cubic-frame construction in section 2. Every member has generic degree three and is nonproper. Our main theorem classifies the entire two-parameter family.

Theorem 1.1 (Complete stable classification). *The stable polynomial left–right orbit set of the family $\{G_{\alpha, \beta}\}$ is*

$$\{\mathcal{O}_0\} \sqcup \{\mathcal{O}_q : q \in \mathbb{C}\},$$

where

$$\mathcal{O}_0 = \{G_{0, \beta} : \beta \in \mathbb{C}\}$$

and

$$\mathcal{O}_q = \{G_{\alpha, \beta} : \alpha \neq 0, \beta/\alpha^2 = q\}.$$

More explicitly:

- (i) all maps $G_{0, \beta}$ are ordinarily left–right equivalent;
- (ii) no $G_{0, \beta}$ is stably equivalent to a member with $\alpha \neq 0$; and
- (iii) if $\alpha, \alpha' \neq 0$, then

$$G_{\alpha, \beta} \sim_{\text{stable}} G_{\alpha', \beta'} \iff \frac{\beta}{\alpha^2} = \frac{\beta'}{\alpha'^2}.$$

The weighted-scaling invariance of β/α^2 is immediate. The content of theorem 1.1 is completeness under *arbitrary* polynomial left–right equivalence and after arbitrary affine stabilization.

Our invariant comes from the set at which the map is not proper. For a dominant equidimensional polynomial map this set, if nonempty, is a hypersurface by Jelonek’s theorem [4]. In our normalized one-parameter slice it has a singular discriminant component D_q and a plane component P . The normalization of D_q is an affine plane containing two intrinsic marked curves:

$$L_q : 3c(c+1)t + qc^2 - 4c - 2 = 0, \quad M : c+1 = 0.$$

The pair (L_q, M) remembers q , even after taking a product with affine space. This gives the necessity in theorem 1.1.

The existence of left–right moduli for nonproper étale maps on other affine varieties has important precedent: Dubouloz and Palka constructed high-dimensional families on affine pseudo-planes [2]. Our theorem concerns polynomial self-maps of affine space arising directly from the new three-dimensional construction. The general marked-root framework and its unstabilized equivalence criterion are developed in the companion paper [5]. The proof below includes all of the special geometry needed for stable separation in the quadratic slice.

2. THE QUADRATIC CUBIC-FRAME FAMILY

Put

$$c = 2x - 3x^2y - x^3z, \quad t = y + \frac{1}{x}, \quad r = \frac{2}{x}. \quad (2.1)$$

For $A = A_\alpha$ and $B = B_{\alpha,\beta}$, let

$$h(T, C) = A(C)T^3 + B(C)T^2$$

and define

$$b = r - \frac{\partial h}{\partial T}(t, c), \quad 2a = h(t, c) + tb. \quad (2.2)$$

Although (2.1) uses $1/x$, the two expressions in (2.2) are polynomials. We write

$$G_{\alpha,\beta} = (a, b, c) : \mathbb{A}_{x,y,z}^3 \longrightarrow \mathbb{A}_{a,b,c}^3.$$

Proposition 2.1. *For every $\alpha, \beta \in \mathbb{C}$, the map $G_{\alpha,\beta}$ is polynomial,*

$$\det DG_{\alpha,\beta} = -2,$$

and its generic degree is three.

Proof. Set

$$\xi = 1 + xy, \quad d = 2 - 3xy - x^2z,$$

so that $t = \xi/x$ and $c = xd$. Clearing denominators in b gives

$$b = \frac{2 + 4\xi - 3d\xi^2}{x} - 3\alpha d^2\xi^2 + 8\alpha d\xi - 2\beta xd^2\xi.$$

The numerator of the displayed fraction vanishes at $x = 0$, hence is divisible by x . Similarly,

$$2a = \frac{2\xi - 2d\xi^3 + 2\xi^2}{x^2} + \frac{4\alpha d\xi^2 - 2\alpha d^2\xi^3}{x} - \beta d^2\xi^2.$$

The first numerator is divisible by x^2 , and the second by x ; this follows directly on substituting $\xi = 1 + xy$ and $d = 2 - 3xy - x^2z$. Thus a and b are polynomial.

On $x \neq 0$, the coordinate changes in (2.1) give

$$\det \frac{\partial(a, b, c)}{\partial(t, r, c)} = \frac{r}{2}, \quad \det \frac{\partial(t, r, c)}{\partial(x, y, z)} = -2x.$$

Since $r = 2/x$, their product is -2 . Polynomiality extends this identity across $x = 0$.

For a target point (a, b, c) , introduce

$$P_{a,b,c}(T) = A(c)T^3 + B(c)T^2 + bT - 2a. \quad (2.3)$$

The construction gives

$$P_{G_{\alpha,\beta}(x,y,z)}(t) = 0, \quad P'_{G_{\alpha,\beta}(x,y,z)}(t) = \frac{2}{x}.$$

Conversely, every finite simple root t reconstructs

$$x = \frac{2}{P'(t)}, \quad y = t - \frac{P'(t)}{2}, \quad z = \frac{2x - 3x^2y - c}{x^3}. \quad (2.4)$$

A general cubic (2.3) has three finite simple roots, giving three source points. $\square$

Remark 2.2. The determinant in proposition 2.1 is -2 , rather than 1, only because of our target normalization. A linear target rescaling gives the usual Keller normalization.

2.1. The ordinary scaling orbit. Suppose $\alpha, \alpha' \neq 0$, and put

$$u = \frac{\alpha}{\alpha'}.$$

If

$$\frac{\beta}{\alpha^2} = \frac{\beta'}{\alpha'^2},$$

then direct substitution gives

$$G_{\alpha',\beta'}\left(ux, \frac{y}{u}, \frac{z}{u^2}\right) = \left(\frac{a}{u^2}, \frac{b}{u}, uc\right) \quad \text{when} \quad (a, b, c) = G_{\alpha,\beta}(x, y, z). \quad (2.5)$$

Thus equality of $q = \beta/\alpha^2$ is sufficient for ordinary equivalence. Every nonzero- α orbit therefore has the normalized representative

$$G_q := G_{1,q},$$

with

$$A(c) = c(c+1), \quad B_q(c) = qc^2 - 4c - 2. \quad (2.6)$$

3. THE INTRINSIC NONPROPERNESS DIVISOR

For a dominant polynomial map F , write S_F for its reduced nonproperness set. Polynomial source automorphisms leave S_F unchanged, target automorphisms carry it isomorphically to the new nonproperness set, and

$$S_{F \times \text{id}_{\mathbb{A}^m}} = S_F \times \mathbb{A}^m. \quad (3.1)$$

The last identity follows either from the valuative criterion or directly from sequences escaping to infinity.

The discriminant of the normalized fiber cubic

$$P_{a,b,c}(T) = A(c)T^3 + B_q(c)T^2 + bT - 2a$$

is

$$\Delta_q = B_q^2 b^2 - 4Ab^3 + 8B_q^3 a - 108A^2 a^2 - 36AB_q ab. \quad (3.2)$$

For $q \neq -2$, let

$$D_q = V(\Delta_q).$$

For $q = -2$, the polynomial Δ_{-2} has content $c + 1$; in that case put

$$D_{-2} = V\left(\frac{\Delta_{-2}}{c+1}\right).$$

In every case let

$$P = V(c+1).$$

Proposition 3.1. *For every $q \in \mathbb{C}$,*

$$S_{G_q} = D_q \cup P.$$

These are the two irreducible components. For $q \neq -2$, the component D_q is singular and P is smooth.

Proof. For $q \neq -2$, the polynomials A and B_q are coprime. Regarding Δ_q as a quadratic in a , one computes

$$\text{Disc}_a(\Delta_q) = 64(B_q^2 - 3Ab)^3.$$

The right side is not a square in $\mathbb{C}(b, c)$, so Gauss's lemma gives the irreducibility of Δ_q .

Use instead the projective root incidence

$$\overline{X}_q = V(AU^3 + B_q U^2 V + bUV^2 - 2aV^3) \subset \mathbb{A}^3 \times \mathbb{P}^1.$$

The source is its simple-root locus with the deleted infinity section over $c = -1$ removed. The simple infinity root over $c = 0$ is retained: it is the image of $x = 0$, and the restriction there is a triangular isomorphism in (y, z) . The incidence is proper, so the image of its boundary is exactly the union of the projective discriminant and the deleted plane P . This proves the assertion for $q \neq -2$, including the retained infinity-root case. See appendix D for the general construction.

At $q = -2$, one has $B_{-2} = -2(c+1)^2$. Dividing the discriminant by its coefficient content $c+1$ gives the unique primitive nonplane component, while the same lost-root argument gives P . Irreducibility of the primitive

discriminant follows by applying the preceding quadratic argument over $\mathbb{C}(c)$ and then Gauss's lemma. This is also the special case of the all-multiplicity nonproperness theorem in [5].

Finally, for $q \neq -2$ the normalization calculation in proposition 4.1 shows that D_q is singular, while P is a plane. $\square$

4. NORMALIZATION AND THE MARKED PAIR

Assume throughout this section that $q \neq -2$. A repeated root t of the inverse cubic gives

$$\begin{aligned} b &= -3A(c)t^2 - 2B_q(c)t, \\ a &= -A(c)t^3 - \frac{1}{2}B_q(c)t^2. \end{aligned} \tag{4.1}$$

Let

$$\nu_q: \mathbb{A}_{c,t}^2 \longrightarrow D_q$$

be the map (4.1), and put

$$H_q(c, t) = 3c(c+1)t + qc^2 - 4c - 2. \tag{4.2}$$

Proposition 4.1. *The map ν_q is the normalization of D_q . Moreover,*

$$\nu_q^{-1}(\text{Sing } D_q) = L_q := V(H_q).$$

Proof. On the parametrization, t satisfies

$$3At^2 + 2B_q t + b = 0, \tag{4.3}$$

$$B_q t^2 + 2bt - 6a = 0. \tag{4.4}$$

Because $\gcd(A, B_q) = 1$, choose $\rho, \sigma \in \mathbb{C}[c]$ with $\rho A + \sigma B_q = 1$. The sum of $\rho/3$ times (4.3) and σ times (4.4) is a monic quadratic equation for t over $\mathbb{C}[D_q]$. Hence ν_q is finite.

Direct substitution gives

$$B_q^2 - 3Ab = H_q^2, \quad -18Aa - B_q b = 2tH_q^2. \tag{4.5}$$

Thus t is rationally recovered away from $H_q = 0$, so ν_q is birational. Since $\mathbb{A}^2$ is normal, it is the normalization.

The pullbacks of the three partial derivatives of Δ_q are

$$(\Delta_q)_a \circ \nu_q = 8H_q^3, \tag{4.6}$$

$$(\Delta_q)_b \circ \nu_q = -4tH_q^3, \tag{4.7}$$

$$(\Delta_q)_c \circ \nu_q = -4t^2(2cq + 2ct + t - 4)H_q^3. \tag{4.8}$$

The first equality shows that all three vanish simultaneously exactly on $H_q = 0$. $\square$

The plane component P marks a second curve upstairs. Write

$$d = q + 2.$$

On $c = -1$, formulas (4.1) become

$$a = -\frac{d}{2}t^2, \quad b = -2dt. \quad (4.9)$$

Since $d \neq 0$, the inverse image of $D_q \cap P$ is the line

$$M := V(c+1) \subset \mathbb{A}_{c,t}^2,$$

and $M \rightarrow D_q \cap P$ is an isomorphism. Notice also that

$$H_q|_M = q+2 \neq 0,$$

so $L_q \cap M = \emptyset$.

The intrinsic data recovered from S_{G_q} are therefore

$$(\mathbb{A}_{c,t}^2, L_q, M). \quad (4.10)$$

The next lemma is the heart of the stable argument.

Lemma 4.2 (Rigidity of the marked cylinder). *Let $q, q' \neq -2$ and $m \geq 0$. Suppose a polynomial automorphism of $\mathbb{A}^{2+m}$ carries*

$$L_q \times \mathbb{A}^m \text{ to } L_{q'} \times \mathbb{A}^m \quad \text{and} \quad M \times \mathbb{A}^m \text{ to } M \times \mathbb{A}^m.$$

Then $q = q'$.

Proof. Let C, T be the pullbacks of the first two coordinates on the target cylinder. Since the prime principal divisor $M \times \mathbb{A}^m$ is preserved, unique factorization gives

$$C+1 = \lambda(c+1) \quad (4.11)$$

for some $\lambda \in \mathbb{C}^*$. For $q \neq -2$, the polynomial H_q is primitive and linear in t , hence irreducible. Preservation of the other prime divisor similarly gives

$$H_{q'}(C, T) = \kappa H_q(c, t) \quad (4.12)$$

for some $\kappa \in \mathbb{C}^*$.

The right side of (4.12) is independent of the stabilization variables and linear in t . Since C is the nonconstant polynomial in (4.11), it follows that

$$T = \mu(c)t + h(c)$$

for polynomials $\mu, h \in \mathbb{C}[c]$. Comparing the coefficients of t in (4.12) and cancelling $c+1$ gives

$$\lambda(\lambda(c+1)-1)\mu(c) = \kappa c. \quad (4.13)$$

The linear polynomial $\lambda(c+1)-1$ must divide c . Therefore $\lambda = 1$, and (4.13) gives $\mu = \kappa$.

The constant term of (4.12) is now

$$3c(c+1)h(c) + B_{q'}(c) = \kappa B_q(c). \quad (4.14)$$

At $c = 0$, both B_q and $B_{q'}$ equal -2 , so $\kappa = 1$. At $c = -1$, equation (4.14) gives $q' + 2 = q + 2$. $\square$

Theorem 4.3. *If $q, q' \neq -2$, then G_q and $G_{q'}$ are stably polynomially left-right equivalent if and only if $q = q'$.*

Proof. Sufficiency is immediate. For necessity, suppose the stabilizations by $\mathbb{A}^m$ are left-right equivalent. By (3.1), the target automorphism identifies

$$(D_q \cup P) \times \mathbb{A}^m \quad \text{with} \quad (D_{q'} \cup P) \times \mathbb{A}^m.$$

The plane component is smooth and the discriminant component is singular, so the two components cannot be exchanged. We obtain an isomorphism

$$D_q \times \mathbb{A}^m \simeq D_{q'} \times \mathbb{A}^m$$

that preserves the intersection with the plane component.

Normalization commutes with adjoining polynomial variables. The isomorphism therefore lifts uniquely to an automorphism of $\mathbb{A}^2 \times \mathbb{A}^m$. The inverse image of the singular locus is $L_q \times \mathbb{A}^m$, and the inverse image of the plane intersection is $M \times \mathbb{A}^m$. The lifted automorphism preserves this marked pair, so lemma 4.2 gives $q = q'$. $\square$

5. THE EXCEPTIONAL VALUE

At $q = -2$,

$$B_{-2}(c) = -2(c+1)^2$$

and Δ_{-2} acquires the factor $c+1$. Let

$$\Delta_{-2}^{\text{res}} = \frac{\Delta_{-2}}{c+1}, \quad D_{-2} = V(\Delta_{-2}^{\text{res}}).$$

Proposition 5.1. *The map G_{-2} is not stably left-right equivalent to G_q for any $q \neq -2$.*

Proof. On the plane $c = -1$, exact substitution gives

$$\Delta_{-2}^{\text{res}}|_{c=-1} = 4b^3.$$

Thus the reduced intersection $D_{-2} \cap P$ is the line $b = 0$. At the point

$$p = (a, b, c) = (0, 0, -1),$$

all three first partial derivatives of Δ_{-2}^{res} vanish. Hence

$$\text{Sing}(D_{-2}) \cap P \neq \emptyset.$$

For $q \neq -2$, by contrast,

$$\left. \frac{\partial \Delta_q}{\partial a} \right|_{c=-1} = 8(q+2)^3 \neq 0.$$

Consequently D_q is smooth along its entire intersection with P , and

$$\text{Sing}(D_q) \cap P = \emptyset.$$

Whether the singular locus of the unique nonplane component meets the plane component is intrinsic to the reduced nonproperness divisor. The same is true after taking a product with affine space. Equation (3.1) therefore separates G_{-2} from every other G_q stably. $\square$

Corollary 5.2. *For all $q, q' \in \mathbb{C}$,*

$$G_q \sim_{\text{stable}} G_{q'} \iff q = q'.$$

Proof. Combine theorem 4.3 and proposition 5.1. $\square$

6. THE LINE $\alpha = 0$ IS GAUGE

It remains to understand the parameter line not covered by the normalization $\alpha = 1$. When $\alpha = 0$,

$$A(C) = C, \quad B(C) = -2 + \beta C^2.$$

Put

$$s = \frac{\beta}{3}, \quad c(x, y, z) = 2x - 3x^2y - x^3z,$$

and define

$$\Theta_s(x, y, z) = \left(x, y + sc, z - 3s\frac{c}{x} \right). \quad (6.1)$$

This is polynomial because

$$\frac{c}{x} = 2 - 3xy - x^2z.$$

It preserves c , has Jacobian determinant one, and has inverse Θ_{-s} .

Set

$$\ell_s(c) = 3s^2c^3 - 4sc, \quad \delta_s(c) = s^3c^4 - 2s^2c^2, \quad (6.2)$$

and define the triangular target automorphism

$$\Xi_s(a, b, c) = \left(a - \frac{1}{2}\delta_s(c) - \frac{sc}{2}b, b + \ell_s(c), c \right). \quad (6.3)$$

Proposition 6.1. *For every $\beta \in \mathbb{C}$,*

$$G_{0,\beta} = \Xi_{\beta/3} \circ G_{0,0} \circ \Theta_{\beta/3}.$$

In particular, the entire line $\alpha = 0$ is one ordinary left-right orbit.

Proof. Because Θ_s fixes c , it sends the framed root $t = y + 1/x$ to $t + sc$. Substitute this translation in

$$cT^3 - 2T^2$$

and compare the coefficients of T with

$$cT^3 + (-2 + 3sc^2)T^2.$$

The residual linear and constant terms are precisely the target shears ℓ_s and δ_s in (6.2). Substitution in (2.2) gives the displayed identity. All expressions are polynomial, and Θ_s, Ξ_s are triangularly invertible. $\square$

Proposition 6.2. *No member with $\alpha = 0$ is stably left-right equivalent to a member with $\alpha \neq 0$.*

Proof. For $\alpha = 0$, the polynomial $A(C) = C$ has only the marked root. The reduced nonproperness set therefore consists solely of its irreducible primitive discriminant component. For $\alpha \neq 0$, the second root $-1/\alpha$ contributes a plane component in addition to the primitive discriminant component, as in proposition 3.1. Thus the two nonproperness sets have respectively one and two irreducible components. Products with $\mathbb{A}^m$ preserve this component count, and (3.1) makes it a stable left-right invariant. $\square$

Proof of theorem 1.1. The first assertion is proposition 6.1; the second is proposition 6.2. For nonzero α, α' , equality of β/α^2 gives the ordinary equivalence (2.5). Conversely, normalize both maps to $\alpha = 1$ using the same formula and apply corollary 5.2. $\square$

7. EXACT VERIFICATION AND PROOF BOUNDARY

The accompanying verifier uses exact SymPy arithmetic. It checks:

- (1) the discriminant (3.2) and its discriminant as a quadratic in a ;
- (2) the normalization formulas (4.1);
- (3) both rational-recovery identities (4.5);
- (4) all three gradient factorizations (4.6)–(4.8);
- (5) the marked-line formulas (4.9);
- (6) the factorization and singular-incidence calculation at $q = -2$;
- (7) the coefficient identity used in lemma 4.2;
- (8) the identity that gauges away β when $\alpha = 0$; and
- (9) the Jacobian identity $\det D\Theta_s = 1$.

The unchanged recovered verifier was rerun in a fresh environment with SymPy declared explicitly. It printed

ALL q-ORBIT CHECKS PASSED,

and regenerated a JSON audit byte-for-byte identical to the recovered reference output.

The principal verifier is `code/main-modulus/verify_q_full_orbit.py`. The companion source also includes separate exact checks for the reciprocal family, the grading-preserving and based moduli problems, conductor arrangements, weighted-lift critical-root invariants, and boundary-residue constructions. They are indexed by mathematical role in `COMPUTATION.md`; their purpose is to make the displayed invariant formulas reproducible without conflating the narrower framed equivalence problems with the stable left–right theorem in the body.

These computations verify the displayed polynomial identities. They do not replace the geometric arguments that the nonproperness divisor is intrinsic, that normalization lifts an isomorphism, or that the two marked curves are recovered as asserted. Those are conventional proofs in sections 3 and 4. A genuinely independent CAS implementation and expert review of the stable-cylinder argument remain release gates.

8. CONSEQUENCES AND QUESTIONS

Theorem 1.1 gives an affine line of stable left–right orbits at one fixed generic degree, together with one additional degenerate orbit. The modulus is not imposed as part of a framing: it is reconstructed from the intrinsic boundary geometry of the map.

The proof deliberately leaves broader claims outside its scope. In particular, it does not classify all cubic-frame counterexamples, all generic-degree-three Keller maps, or the full stable moduli functor.

Question 8.1. *Which larger finite-dimensional cubic-frame families admit complete stable invariants recovered from their normalized nonproperness divisors?*

Question 8.2. *Can the marked-cylinder rigidity argument be formulated as a general cancellation theorem for pairs $(\mathbb{A}^2, L, M)$?*

Question 8.3. *What scheme or stack represents deformations of the maps in theorem 1.1 modulo stable polynomial left-right equivalence?*

Question 8.4 (Compactification across boundary collisions). *Can the all-multiplicity finite-root classification of theorem D.5 be compactified across strata where deleted roots escape to $c = \infty$? The desired object should retain the weighted relative-Jacobian data while allowing the number of finite marked planes to change.*

APPENDIX A. THE RECIPROCAL CUBIC FAMILY

The family in the body gives a clean affine line of stable orbits. A second construction gives much larger fixed-generic-degree families and boundary curves of arbitrary genus. We record it separately because the algebraic construction is exact, while the last stable-separation step still requires a specialist audit of the birational argument.

Put

$$\begin{aligned} A &= 1 + xy, & B &= A^2z + y^2(4 + 3xy), & P &= AB, \\ Q &= y + 3xB, & R &= 2x - 3x^2y - x^3z. \end{aligned}$$

Thus (P, Q, R) is the unnormalized degree-seven map. On $A \neq 0$, set

$$s = \frac{x}{A}.$$

Then $Ps = xB$ and $1 - ys = A^{-1}$.

A.1. Pole cancellation. Let $g(p) \in \mathbb{C}[p]$. In the rational coordinates (P, s, y) , consider

$$q = y + 3Pg(P)s, \quad r = 2Pg(P)s^3 - qs^2 + 2s.$$

Proposition A.1 (Reciprocal pole-cancellation criterion). *The prescription (P, q, r) extends to a polynomial self-map of $\mathbb{A}^3$ if and only if*

$$g(0) = 1, \quad g'(0) = 0.$$

Equivalently, $g(p) = 1 + p^2\eta(p)$ for a unique $\eta \in \mathbb{C}[p]$.

Proof. The second coordinate is $q = y + 3xBg(P)$. Comparing the third coordinate with R gives

$$r - R = -\frac{x^3B}{A^2}(g(P) - 1).$$

Write $g(P) - 1 = c_0 + c_1P + P^2\eta(P)$. Since $\gcd(A, x^3B) = 1$ in $\mathbb{C}[x, y, z]$, polynomiality forces $A^2 \mid c_0 + c_1AB$. Reducing first modulo A , and then after one division by A , gives $c_0 = c_1 = 0$. The converse follows from

$$r - R = -x^3B^3\eta(P).$$

□

For $\eta \in \mathbb{C}[p]$, define

$$\begin{aligned} Q_\eta &= Q + 3xA^2B^3\eta(P), \\ R_\eta &= R - x^3B^3\eta(P), \\ \mathcal{G}_\eta &= (P, Q_\eta, R_\eta), \end{aligned} \tag{A.1}$$

and write $g(p) = 1 + p^2\eta(p)$.

Theorem A.2 (Algebraic properties of the reciprocal family). *For every η ,*

$$\det J\mathcal{G}_\eta = -2.$$

The map has generic degree three and generic Galois group S_3 .

Proof. In rational coordinates,

$$Q_\eta = y + 3Pg(P)s, \quad R_\eta = 2Pg(P)s^3 - Q_\eta s^2 + 2s.$$

Direct differentiation gives

$$\det \frac{\partial(P, Q_\eta, R_\eta)}{\partial(P, s, y)} = -\frac{2}{A}, \quad \det \frac{\partial(P, s, y)}{\partial(x, y, z)} = A.$$

Their product is -2 .

For target coordinates (p, q, r) , the primitive element satisfies

$$\Omega_{p,q,r}(S) = 2pg(p)S^3 - qS^2 + 2S - r = 0. \tag{A.2}$$

Moreover $y = q - 3pg(p)s$, so $\mathbb{C}(x, y, z) = \mathbb{C}(p, q, s)$. Over $\mathbb{C}(p, q)$, the right side defining r is a degree-three rational function of s , proving generic degree three and irreducibility. The discriminant is $4\Delta_\eta$, where

$$\Delta_\eta = q^2 - 16pg - q^3r + 18pgqr - 27p^2g^2r^2.$$

As a quadratic in r , its discriminant is $(q^2 - 12pg)^3$, which is not a square in the generic function field. The irreducible cubic therefore has Galois group S_3 . □

A.2. The image and finite completion. Define

$$\Gamma_\eta : \quad q^2 = 12pg(p), \quad 3qr = 4,$$

and

$$\Lambda_\eta : \quad g(p) = 0, \quad qr = 1.$$

Theorem A.3 (Exact image). *Set-theoretically over $\mathbb{C}$,*

$$\mathbb{A}^3 \setminus \mathcal{G}_\eta(\mathbb{A}^3) = \Gamma_\eta \sqcup \Lambda_\eta.$$

Proof. A simple root s of (A.2) reconstructs a unique source point. Indeed, with

$$D = \Omega'(s), \quad a = \frac{2}{D}, \quad x = as, \quad y = q - 3pg(p)s, \quad b = \frac{p}{a},$$

put

$$z = \frac{b - y^2(4 + 3xy)}{a^2}.$$

Then $1 + xy = a$, $B = b$, and substitution gives the target (p, q, r) .

For $p \neq 0$, a target is therefore omitted precisely when Ω has no simple root. If $g(p) \neq 0$, this happens exactly when the cubic is a cube of a linear polynomial, which gives Γ_η . If $g(p) = 0$, the equation is quadratic and has no simple root exactly when $qr = 1$, which gives Λ_η .

The plane $p = 0$ is in the image. If $p = q = 0$, take $(x, y, z) = (r/2, 0, 0)$. If $p = 0$ and $q \neq 0$, take

$$x = \frac{2}{q}, \quad y = -\frac{q}{2}, \quad z = \frac{5q^2}{4} - \frac{\eta(0)q^6}{64} - \frac{rq^3}{8}.$$

Neither displayed omitted locus meets this plane. $\square$

Let

$$\mathcal{X}_\eta = V(2pg(p)u^3 - qu^2v + 2uv^2 - rv^3) \subset \mathbb{A}_{p,q,r}^3 \times \mathbb{P}_{[u:v]}^1.$$

The projection $\mathcal{X}_\eta \rightarrow \mathbb{A}^3$ is finite flat of rank three. The source embeds by

$$(x, y, z) \mapsto (\mathcal{G}_\eta(x, y, z), [x : A]).$$

If $\mathcal{R}_\eta$ is the repeated-root divisor and $\mathcal{I}_\eta = V(v, g(p))$, the reconstruction above in the two projective charts gives

$$\mathbb{A}^3 \simeq \mathcal{X}_\eta \setminus (\mathcal{R}_\eta \cup \mathcal{I}_\eta)$$

and hence

$$S_{\mathcal{G}_\eta}^{\text{red}} = V(g(p)\Delta_\eta(p, q, r)). \quad (\text{A.3})$$

This finite completion is also the natural frame for the moduli assertions below.

A.3. Curves and stable separation. The component Γ_η is a dense open of the normalization of

$$C_\eta : u^2 = 12p(1 + p^2\eta(p)).$$

If $\eta \neq 0$ has degree d and the polynomial on the right is squarefree, then

$$g(C_\eta) = \left\lfloor \frac{d+2}{2} \right\rfloor.$$

Each distinct root of g contributes a rational $\mathbb{G}_m$ -component to Λ_η .

Theorem A.4 (Stable curve invariant). *Assume that C_η and C_θ are smooth projective curves of positive genus. If $\mathcal{G}_\eta$ and $\mathcal{G}_\theta$ are polynomially left-right equivalent after stabilization, then*

$$C_\eta \simeq C_\theta.$$

Consequently the reciprocal family contains positive-dimensional families of fixed generic degree. Their boundary curves realize every positive genus.

Proof. Stable left-right equivalence identifies the reduced nonproperness loci after taking a product with affine space. In (A.3), the components arising from Λ_η are rational, whereas Γ_η has projective normalization C_η . Hence an isomorphism of stabilized nonproperness loci must carry the unique nonrational component to the unique nonrational component. On function fields it gives

$$\mathbb{C}(C_\eta)(t_1, \dots, t_m) \simeq \mathbb{C}(C_\theta)(s_1, \dots, s_m).$$

Thus C_η and C_θ are stably birational. Smooth projective curves of positive genus are the maximal rationally connected quotients of their products with projective space, so stable birationality forces C_η and C_θ to be birational. Smooth projective birational curves are isomorphic.

For each genus, take

$$g_{\tau,m}(p) = (1 - p^2)(1 - p^2/\tau^2) \prod_{j=3}^m (1 - p^2/\alpha_j^2).$$

Varying the symmetric branch set gives nonisomorphic curves for general τ . Moreover, $\eta = (g-1)/p^2$ is polynomial and the branch cross-ratios vary. $\square$

A.4. The collision-preserving line. For

$$\eta_\lambda(p) = \frac{\lambda}{3}(4p+1), \quad g_\lambda(p) = 1 + \frac{\lambda}{3}p^2(4p+1),$$

the projective normalization is the short Weierstrass curve with

$$j(\lambda) = \frac{1728\lambda}{\lambda + 324}.$$

For $\lambda \neq 0, -324$, the omitted set has one elliptic component and three rational components. At $\lambda = 0$, it has one component $\mathbb{G}_m$. At $\lambda = -324$,

$$g_{-324}(p) = -(6p+1)^2(12p-1),$$

and the reduced omitted locus has three components: two copies of $\mathbb{G}_m$ and a rational curve whose normalization is $\mathbb{P}^1$ minus six points. By theorem A.4, these data give a complete pairwise separation of the λ -line.

A.5. Degree and framed moduli. If

$$g(p) = 1 + c_2 p^2 + \cdots + c_m p^m, \quad c_m \neq 0,$$

then

$$\deg \mathcal{G}_\eta = 7m + 6.$$

Thus degree 20 is the first nonbase member of this reciprocal normal form, while degree 27 is the first positive-dimensional quotient. These are normal-form statements, not absolute lower bounds under arbitrary polynomial equivalence. The separate coarse argument presently gives only

$$4 \leq d_{\text{positive genus}} \leq 20.$$

Fix a degree bound m for η and retain the finite completion, infinity section, and target coordinates as part of the framing. Let

$$V_m = \{\eta \in \mathbb{C}[p] : \deg \eta \leq m\}.$$

The residual source scaling acts on the coefficient of p^i with weight $-(2i+4)$. Exact pole cancellation and a calculation of frame-preserving source–target transformations give the framed moduli groupoid

$$[V_m / \mathbb{G}_m].$$

Its tangent complex at η is the standard two-term complex

$$\mathrm{Lie}(\mathbb{G}_m) \longrightarrow T_\eta V_m.$$

With the boundary marked, differentiation of (A.2) gives the corresponding logarithmic deformation complex. This is a statement about the framed reciprocal family; it is not a presentation of the full unframed Keller moduli stack.

APPENDIX B. ADDITIONAL RESTRICTED MODULI RESULTS

The results in this appendix concern narrower equivalence problems than the stable left–right problem solved in the body. They are retained because they identify useful intermediate moduli spaces, but they should not be read as classifications of all Keller maps of the stated degree.

B.1. A grading-preserving degree-five modulus. For the mixed grading

$$\deg(x, y, z) = (-1, 1, 2),$$

the equivariant Keller equation reduces to a finite coefficient system in the quotient variables $u = xy$ and $v = x^2 z$. If

$$X = xf(u, v), \quad Y = x^{-1}g(u, v), \quad Z = x^{-2}h(u, v),$$

that system is

$$gJ(f, h) - 2hJ(f, g) + fJ(g, h) = 1.$$

Proposition B.1 (Restricted fixed-degree moduli). *Inside the analyzed generic-degree-five equivariant family, the grading-preserving left–right orbit space contains an affine one-parameter family. Its members are separated by the affine-scaling class of the critical-root configuration of the associated one-variable polynomial.*

Exact-elimination proof record. The weight restrictions leave finitely many coefficients. Solving the determinant equations gives a family whose remaining data include a one-variable critical polynomial. Every grading-preserving source and target transformation acts on its roots through the restricted affine scaling group. A cross-ratio-type invariant of the critical roots varies, which separates the displayed one-parameter subfamily. The archived elimination and transformation calculation is exact; the proposition makes no assertion about equivalences that do not preserve the grading. $\square$

B.2. The low-degree cubic frame and its based parameter.

Proposition B.2 (Based low-degree normal form). *In the reciprocal boundary stratum of ordinary degree at most seven, the pole-cancellation and Keller equations reduce the cubic frame to a normal form with a single residual coefficient κ when the boundary frame and basepoint are fixed. Residual scaling gives exactly two based classes, $\kappa = 0$ and $\kappa \neq 0$. After the frame is forgotten, a determinant-one source–target translation moves κ , so it is not an invariant of ordinary left–right equivalence.*

Proof. Solving the coefficient equations in the fixed reciprocal frame eliminates every coefficient except κ . The residual diagonal action is transitive on its nonzero values. Substitution of the explicit determinant-one translation into the cubic-frame formula changes κ while preserving the unframed map class. $\square$

The pole-cancellation equations in the general cubic marked-root frame admit an exact normal form. In ordinary degree at most seven, every member of that frame is polynomially source–target equivalent to the base map. Thus the frame itself produces no additional unbased degree-seven modulus.

If the boundary frame and basepoint are fixed, a residual coefficient traditionally denoted κ remains. The exact restricted classification has two based classes:

$$\kappa = 0, \quad \kappa \neq 0.$$

The nonzero values are related by the residual scaling. On forgetting the frame, the literal coefficient can be moved by a determinant-one source–target translation; it is therefore not an invariant of ordinary left–right equivalence.

Coefficientwise formal trivializations of the nonzero based family exist by formal étaleness, but their spatial degrees grow with the deformation order. The recovered calculation gives componentwise degrees

$$(36, 34, 36), \quad (71, 69, 71)$$

for the first two recursively determined coefficients. These values are evidence of growth, not a lower-bound theorem for every possible trivialization.

Remark B.3 (Why these statements are separate). Proposition B.1 uses a grading as part of the equivalence problem. The κ -classification uses a fixed boundary frame and basepoint. Neither competes with the intrinsic stable invariant q proved in the body, nor with the reciprocal-family boundary invariant of appendix A.

APPENDIX C. BOUNDARY RESIDUES FOR GENERAL SQUAREFREE CUBIC FRAMES

The one-parameter invariant is the first member of a general construction. We record the squarefree statement here; the full standalone sequel source, including expanded calculations and examples, is retained under [supplements/](#).

Definition C.1 (Admissible frame). A pair $A, B \in \mathbb{C}[c]$ is admissible if

$$A(0) = 0, \quad A'(0) = 1, \quad B(0) = -2, \quad B'(0) = -2A''(0).$$

Using

$$c = 2x - 3x^2y - x^3z, \quad t = y + \frac{1}{x}, \quad r = \frac{2}{x},$$

define $G_{A,B} = (a, b, c)$ by

$$b = r - 3A(c)t^2 - 2B(c)t, \quad 2a = A(c)t^3 + B(c)t^2 + tb.$$

Proposition C.2 (Universal cubic frame). *For every admissible pair, $G_{A,B}$ is polynomial and*

$$\det DG_{A,B} = -2.$$

If $A \neq 0$, its generic degree is three.

Proof. The admissibility jets cancel the possible poles at $x = 0$. On $x \neq 0$, the change $(x, y, z) \mapsto (t, r, c)$ has determinant $-2x$, while $(t, r, c) \mapsto (a, b, c)$ has determinant $r/2$. Since $r = 2/x$, the product is -2 , and polynomiality extends it across $x = 0$. Finite simple roots of

$$A(c)T^3 + B(c)T^2 + bT - 2a$$

recover the source, giving generic degree three. □

Assume now that A is squarefree and $\gcd(A, B) = 1$. Put

$$Z_A^\circ = V(A/c) \subset \mathbb{G}_m.$$

Homogenizing the inverse cubic gives

$$A(c)U^3 + B(c)U^2V + bUV^2 - 2aV^3 = 0.$$

At a root s of A , in the local coordinate V/U , the transverse derivative of the infinity section is $B(s)$. The root over $c = 0$ is retained in the affine source; the roots in Z_A° are deleted. Thus the natural decorated infinity datum is

$$\left(Z_A^\circ, \sigma_B = B|_{Z_A^\circ} \right).$$

The discriminant component is normalized by

$$\nu_{A,B}: \mathbb{A}_{c,t}^2 \longrightarrow D_{A,B},$$

$$b = -3A(c)t^2 - 2B(c)t, \quad a = -A(c)t^3 - \frac{1}{2}B(c)t^2.$$

The preimage of its singular locus is

$$L_{A,B} = V(3A(c)t + B(c)),$$

and the deleted infinity components lift to the vertical lines

$$M_s = V(c - s), \quad s \in Z_A^\circ.$$

Theorem C.3 (Stable boundary Torelli on the squarefree locus). *Let (A, B) and $(\tilde{A}, \tilde{B})$ be admissible squarefree coprime pairs. The following are equivalent:*

- (i) $G_{A,B}$ and $G_{\tilde{A},\tilde{B}}$ are stably polynomially left-right equivalent;
- (ii) they are ordinarily polynomially left-right equivalent;
- (iii) there is $u \in \mathbb{C}^*$ such that

$$\tilde{A}(uc) = uA(c), \quad \tilde{B}(uc) - B(c) \in cA(c)\mathbb{C}[c];$$

- (iv) multiplication by u identifies the decorated finite schemes

$$(Z_A^\circ, \sigma_B) \quad \text{and} \quad (Z_{\tilde{A}}^\circ, \sigma_{\tilde{B}}).$$

Proof. A stable equivalence preserves the reduced nonproperness divisor. Its unique singular nonplane component is the discriminant component, while its other components are the planes $c = s$, $s \in Z_A^\circ$. Passing to normalizations gives an automorphism of a cylinder over $\mathbb{A}_{c,t}^2$ preserving $L_{A,B}$ and the union of the vertical lines.

Write $Q_A = A/c$, and let C, T be the pullbacks of the normalization coordinates. Unique factorization gives

$$Q_{\tilde{A}}(C) = \lambda Q_A(c).$$

Hence C is algebraic over $\mathbb{C}(c)$. Since $\mathbb{C}(c)$ is algebraically closed in the purely transcendental cylinder function field, $C \in \mathbb{C}(c)$. Polynomiality of the automorphism and its inverse then force

$$C = uc + v.$$

The omitted vertical roots account for every root except the distinguished retained root, so the latter is preserved and $v = 0$. The normalization $A'(0) = \tilde{A}'(0) = 1$ yields

$$\tilde{A}(uc) = uA(c).$$

Preservation of the irreducible singular-preimage curve gives

$$3\tilde{A}(C)T + \tilde{B}(C) = \kappa(3A(c)t + B(c)).$$

The value at $c = 0$ gives $\kappa = 1$, and the admissibility jets imply

$$\tilde{B}(uc) - B(c) \in cA(c)\mathbb{C}[c].$$

This proves (i) $\Rightarrow$ (iii); (iii) and (iv) are equivalent by squarefreeness.

Conversely, after the diagonal change

$$(x, y, z) \mapsto (ux, y/u, z/u^2), \quad (a, b, c) \mapsto (a/u^2, b/u, uc),$$

condition (iii) reduces to

$$\tilde{B} - B = 3A\phi, \quad \phi \in c\mathbb{C}[c].$$

The source root translation

$$(x, y, z) \mapsto \left(x, y + \phi(c), z - 3\frac{\phi(c)}{x} \right)$$

is polynomial, preserves c , and shifts t by $\phi(c)$. Expanding the cubic supplies a triangular target automorphism, proving ordinary left–right equivalence. $\square$

Corollary C.4 (Arbitrarily large fixed-degree stable moduli). *If $\deg A = N + 1$, the generic squarefree orbit space is*

$$\frac{\{(s_i, v_i)_{i=1}^N \in (\mathbb{G}_m \times \mathbb{G}_m)^N : s_i \neq s_j\}}{\mathbb{G}_m \times S_N},$$

where $\mathbb{G}_m$ scales the s_i and fixes the v_i . Its generic dimension is $2N - 1$. Fixing a root configuration with no scaling symmetry leaves an N -dimensional family of pairwise stably inequivalent maps. With the canonical interpolating representative for B , every map in such a family has

$$\begin{aligned} \det DG &= -2, & \mu(G) &= 3, \\ (\deg G_1, \deg G_2, \deg G_3) &= (4N + 7, 4N + 6, 4). \end{aligned}$$

Remark C.5 (The quadratic modulus). For

$$A = c + \alpha c^2, \quad B = -2 - 4\alpha c + \beta c^2,$$

the unique deleted infinity point is $-1/\alpha$, and

$$B(-1/\alpha) = 2 + \frac{\beta}{\alpha^2} = q + 2.$$

Thus q is the normalized transverse derivative at the deleted infinity section. Equivalently, it is the principal part of the rational Tschirnhaus translation needed to identify the cubic frames. At $q = -2$ that derivative vanishes and the deleted section becomes ramified.

Proof boundary. The exact verifier checks all displayed polynomial identities, the frame Jacobian, the marked infinity section, the root-translation gauge, and an explicit three-parameter interpolation family. The four load-bearing geometric steps—identification of the affine open in the finite completion, equality with the nonproperness locus, lifting stable equivalences through normalization, and rigidity of the marked vertical cylinder—still require independent specialist review.

APPENDIX D. ALL MULTIPLICITIES VIA THE RELATIVE JACOBIAN
BLOWUP

The squarefree boundary-residue argument extends to arbitrary common roots once the absolute singular locus is replaced by the relative Jacobian over the intrinsic affine c -line. Let (A, B) be an admissible cubic frame, put $Q = A/c$, and write

$$P_{a,b,c}(T) = A(c)T^3 + B(c)T^2 + bT - 2a.$$

D.1. Projective incidence and the primitive discriminant. Consider

$$\overline{X}_{A,B} = V\left(AU^3 + BU^2V + bUV^2 - 2aV^3\right) \subset \mathbb{A}_{a,b,c}^3 \times \mathbb{P}_{[U:V]}^1.$$

The marked-root map extends regularly by

$$(x, y, z) \longmapsto (G_{A,B}(x, y, z), [1 + xy : x]).$$

If $\overline{X}^{\text{simp}}$ is the simple-projective-root locus, then

$$\mathbb{A}^3 \simeq \overline{X}^{\text{simp}} \setminus V(V, Q(c)). \quad (\text{D.1})$$

On the finite-root chart the inverse is the usual reconstruction formula. At $x = 0$, the retained infinity root lies over $c = 0$; writing

$$A = c + a_2c^2 + a_3c^3 + \cdots, \quad B = -2 - 4a_2c + b_2c^2 + \cdots,$$

its restriction is the triangular map

$$b = y + 4a_2, \quad a = z + 4y^2 + 2a_2y - 2b_2 - 8a_3.$$

Thus the retained infinity section belongs to the affine source.

The cubic discriminant is

$$\Delta = B^2b^2 - 4Ab^3 + 8B^3a - 108A^2a^2 - 36ABab.$$

Proposition D.1 (Primitive discriminant). *Its coefficient content in a, b is*

$$\chi(c) = \gcd(A(c), B(c)^2),$$

normalized by $\chi(0) = 1$. The polynomial $\delta = \Delta/\chi$ is irreducible, and

$$S_{G_{A,B}} = V(\delta) \cup \bigcup_{s \in V(Q)_{\text{red}}} V(c - s).$$

Proof. At s , the minimum coefficient valuation of Δ is

$$\min\{\text{ord}_s A, 2 \text{ord}_s B\},$$

which gives the content formula. Over $\mathbb{C}(c)$,

$$\text{Disc}_a(\Delta) = 64(B^2 - 3Ab)^3,$$

which is not a square in $\mathbb{C}(c, b)$; Gauss's lemma gives irreducibility after removing content. Finally, equation (D.1) identifies the boundary of a proper incidence morphism: its two kinds of images are nonsimple roots and the deleted simple infinity roots. $\square$

Proposition D.2 (Intrinsic common planes). *For a deleted root s of A ,*

$$B(s) = 0 \iff V(\delta) \cap V(c - s) \text{ is scheme-theoretically nonreduced.}$$

Proof. Put $u = c - s$, $A = u^r \alpha$, and $B = u^k \beta$, with the displayed units at $u = 0$. If $k = 0$, the restriction is the reduced smooth conic

$$B(s)^2(b^2 + 8B(s)a).$$

If $k > 0$, its primitive restriction is, according as $r < 2k$, $r = 2k$, or $r > 2k$,

$$-4\alpha(0)b^3, \quad b^2(\beta(0)^2 - 4\alpha(0)b), \quad \beta(0)^2b^2.$$

□

D.2. The relative Gauss chart. The pencil of deleted planes recovers c up to affine change. On $D = V(\delta)$, define

$$\mathfrak{J}_{D/\mathbb{A}^1} = \text{Fitt}_1 \Omega_{D/\mathbb{A}^1} = (\delta_a, \delta_b).$$

Let $\mathfrak{B}_D = \text{Bl}_{\mathfrak{J}} D$. Write

$$g = \gcd(A, B), \quad A = gA_0, \quad B = gB_0,$$

and put

$$H_0 = 3A_0t + B_0, \quad \rho = g^3/\chi.$$

Proposition D.3 (Finite-root chart and weighted divisor). *The δ_a -chart of $\mathfrak{B}_D$ is canonically $\mathbb{A}_{c,t}^2$. On it,*

$$\mathfrak{J}\mathcal{O}_{\mathbb{A}^2} = (\rho(c)H_0(c, t)^3).$$

Proof. Under the repeated-root parametrization

$$b = -3At^2 - 2Bt, \quad a = -At^3 - \frac{1}{2}Bt^2,$$

one obtains

$$\delta_a = 8\rho H_0^3, \quad \delta_b = -4t\rho H_0^3.$$

Hence $t = -2\delta_b/\delta_a$, so adjoining the blowup ratio adjoins exactly t ; conversely a, b are polynomial in c, t . □

This chart is intrinsic. Indeed, on the other chart put $q = \delta_a/\delta_b = -2/t$. Then

$$B(c) = qb + \frac{3}{2}q^2a, \quad A(c) = \frac{1}{4}q^2b + \frac{1}{2}q^3a.$$

The complement $q = 0$ therefore maps precisely into the curve intersections with the common deleted planes characterized by proposition D.2; removing the prime divisors above those curves leaves the chart of proposition D.3.

D.3. Multiplicity recovery and stable Torelli. At a deleted root s , let

$$r = \text{ord}_s A, \quad k = \text{ord}_s B, \quad m = \min(r, k).$$

The intrinsic chart records

$$p = \text{ord}_s A_0 = r - m, \quad d = \text{ord}_s \rho = 3m - \min(r, 2k).$$

Lemma D.4 (Multiplicity recovery). *The common-factor multiplicity is*

$$m = \begin{cases} d, & p \geq d, \\ (p + d)/2, & p < d. \end{cases}$$

Theorem D.5 (All-multiplicity fixed-frame Torelli). *Let (A, B) and $(\tilde{A}, \tilde{B})$ be admissible cubic frames, with A/c and $\tilde{A}/c$ nonconstant. The following are equivalent:*

- (i) $G_{A,B}$ and $G_{\tilde{A},\tilde{B}}$ are stably left–right equivalent;
- (ii) they are ordinarily polynomially left–right equivalent;
- (iii) there is $u \in \mathbb{C}^\times$ such that

$$\tilde{A}(uc) = uA(c), \quad \tilde{B}(uc) - B(c) \in cA(c)\mathbb{C}[c].$$

No squarefreeness or coprimality hypothesis is required.

Proof. The intrinsic finite-root chart, its marked vertical lines, and the divisor $\text{div}(\rho H_0^3)$ all survive a product with affine space. A stable equivalence preserves this data. A marked vertical line forces the pulled-back base coordinate to have the form $C = uc + v$. The unique nonvertical divisor forces

$$3\tilde{A}_0(C)T + \tilde{B}_0(C) = \kappa(3A_0(c)t + B_0(c)).$$

Thus $T = \mu(c)t + h(c)$. The cylinder Jacobian makes μ a nonzero constant and excludes dependence on stabilization variables.

The horizontal divisor gives every p_s , while the vertical multiplicity gives every d_s . By lemma D.4, all multiplicities of A are therefore preserved. The deleted planes account for every root except the retained root zero, so $v = 0$. Comparing the t -coefficient and using the admissible first jets gives

$$\tilde{A}(uc) = uA(c).$$

The constant coefficient then gives

$$\tilde{B}(uc) - B(c) = -3uA(c)h(c).$$

Differentiating at zero shows $h(0) = 0$, which is condition (iii). Conversely, the explicit root translation used in the ordinary classification constructs a polynomial left–right equivalence and does not use squarefreeness or coprimality. $\square$

Equivalently, scaling identifies the decorated Artin schemes

$$(\text{Spec } \mathbb{C}[c]/(A/c), B \bmod (A/c)).$$

Proposition D.6 (The exceptional weighted stratum). *For the exceptional quadratic member $A = c(c + 1)$, $B = -2(c + 1)^2$, one has*

$$\rho = (c + 1)^2, \quad H_0 = 3ct - 2(c + 1).$$

Thus its special behavior is the intrinsic vertical multiplicity two, not an ad hoc extra orbit invariant.

Proof. Here $g = c + 1$, $A_0 = c$, $B_0 = -2(c + 1)$, and $\chi = c + 1$. Substitution in proposition D.3 gives

$$\mathfrak{J}\mathcal{O}_{\mathbb{A}^2} = ((c + 1)^2(3ct - 2(c + 1))^3),$$

so the vertical multiplicity is exactly two. $\square$

Question D.7 (Gluing at infinity). *Can the classification be compactified across strata in which deleted roots escape to $c = \infty$, so that the number of finite marked planes changes? This includes comparison with the $\alpha = 0$ stratum and remains separate from the finite-root theorem above.*

APPENDIX E. FIXED-DEGREE MODULI IN THE WEIGHTED-LIFT CONSTRUCTION

The cubic-frame families studied in the body have generic degree three. There is also a fixed-degree moduli theorem for the public weighted-lift construction. A seed polynomial p of degree d satisfies

$$p(0) = 0, \quad p(1) = -1, \quad \int_0^1 p(w) dw = 0$$

and produces a counterexample of generic degree $n = d + 1$. Put

$$q(w) = wp(w) - \int_0^w p(s) ds, \quad q'(w) = wp'(w).$$

For $d = 4$, the two-dimensional normalized seed family is already explicit:

$$p_{C,D}(w) = \left(2 + \frac{C}{2} + \frac{4D}{5}\right) w \left(-3 - \frac{3C}{2} - \frac{9D}{5}\right) w^2 + Cw^3 + Dw^4.$$

Thus the first nontrivial fixed-degree case is not merely a dimension count. For the cubic $p'_{C,D}$, two invariants under simultaneous rescaling of the polynomial and its variable are

$$I(C, D) = -\frac{9C(5C + 6D + 10)}{2D(5C + 8D + 20)}, \quad J(C, D) = \frac{135C^3}{8D^2(5C + 8D + 20)}.$$

Their Jacobian is

$$\det \frac{\partial(I, J)}{\partial(C, D)} = -\frac{30375C^3}{2D^4(5C + 8D + 20)^3},$$

so they separate a two-parameter family on a dense open set.

Theorem E.1 (Intrinsic critical-root arrangement). *Assume*

$$p'(0) \neq 0, \quad p' \text{ is squarefree with nonzero roots}, \quad \gcd(p, q) = w.$$

The unique nonnormal hypersurface component S_p of the nonproperness set has normalization

$$\nu_p : \mathbb{A}_{u, \zeta}^2 \longrightarrow S_p, \quad \nu_p(u, \zeta) = \left(\frac{q(\zeta u)}{\zeta^2}, \frac{p(\zeta u)}{\zeta}, \zeta \right).$$

Its ramification divisor is

$$R_p = V(p'(\zeta u)) = \bigcup_{r \in \text{Crit}(p)} V(\zeta u - r).$$

The unordered critical-root configuration of p , modulo common scaling, is invariant under arbitrary polynomial left–right equivalence of the weighted-lift maps.

Proof. The displayed quotients are polynomials because $p(0) = 0$ and q vanishes to order at least two. Write $w = \zeta u$. If a valuation ring contains

$$\zeta, \quad p(w)/\zeta, \quad q(w)/\zeta^2,$$

then it contains u : negative valuation of w is incompatible with integrality of $p(w)/\zeta$; positive valuation and $p'(0) \neq 0$ give $v(w) \geq v(\zeta)$; and $v(w) = 0 < v(\zeta)$ would give a nonzero common root of p and q . Thus ν_p is finite. It is birational on the open reconstruction locus, and $\mathbb{A}^2$ is normal, so it is the normalization.

Differentiation gives

$$\partial_u \left(\frac{p(\zeta u)}{\zeta} \right) = p'(\zeta u), \quad \partial_u \left(\frac{q(\zeta u)}{\zeta^2} \right) = u p'(\zeta u),$$

which proves the ramification formula.

All other hypersurface components of the nonproperness set lie in the plane $\zeta = 0$ and are normal. Hence S_p is intrinsically the unique nonnormal component, and an arbitrary left–right equivalence lifts to an automorphism of its normalization carrying R_p to $R_{\tilde{p}}$. Componentwise,

$$\phi^*(u' \zeta' - \tilde{r}_i) = \lambda_i(u \zeta - r_{\sigma(i)}).$$

Subtracting two identities makes all λ_i equal, so

$$\phi^*(u' \zeta') = \lambda u \zeta + \mu.$$

The left side factors into two coordinate polynomials. If $\mu \neq 0$, $\lambda u \zeta + \mu$ is irreducible, a contradiction. Thus $\mu = 0$, and

$$\{\tilde{r}_i\} = \lambda \{r_i\}.$$

□

Corollary E.2 (Moduli at every fixed generic degree). *For every $n \geq 4$, the weighted-lift construction contains an $(n-3)$ -dimensional family of pairwise inequivalent counterexamples of generic degree n , even under arbitrary polynomial left–right equivalence.*

Proof. The degree- $d = n - 1$ seed space cut out by the three displayed linear conditions has dimension $d - 2 = n - 3$. The map to critical-root configurations modulo scaling is generically finite. Indeed, after fixing a representative $\{r_i\}$, every derivative has the form

$$p'(w) = K \prod_{i=1}^{d-1} (w - \lambda r_i).$$

The integral condition becomes

$$\int_0^1 (1-s) \prod_i (s - \lambda r_i) ds = 0.$$

This is a nonzero polynomial equation in λ ; at $\lambda = 0$, its value is $1/(d(d+1))$. There are therefore only finitely many λ , and $p(1) = -1$ then determines K . Apply theorem E.1. $\square$

Remark E.3. The dimension statement means that the seed variety maps generically finitely to algebraically separating invariants. It does not assert the existence of a separated coarse moduli scheme.

APPENDIX F. EXACT CONDUCTORS AND GROWING STABLE MODULI

We return to the cubic frame with

$$h(T, C) = A(C)T^3 + B(C)T^2.$$

The target fiber cubic is

$$P(T) = A(c)T^3 + B(c)T^2 + vT - 2u,$$

with discriminant

$$\Delta_{A,B} = B^2v^2 - 4Av^3 + 8B^3u - 108A^2u^2 - 36ABuv.$$

Assume first that $\gcd(A, B) = 1$.

F.1. The conductor.

Theorem F.1 (Exact discriminant conductor). *The map*

$$\nu : \mathbb{A}_{c,t}^2 \longrightarrow V(\Delta_{A,B}),$$

$$v = -3A(c)t^2 - 2B(c)t, \quad u = -A(c)t^3 - \frac{1}{2}B(c)t^2$$

is the normalization. If

$$H(c, t) = 3A(c)t + B(c),$$

then the conductor in $\mathbb{C}[c, t]$ is exactly

$$H^2\mathbb{C}[c, t].$$

Its contraction to the discriminant coordinate ring is generated by

$$B^2 - 3Av, \quad -18Au - Bv.$$

Proof. Irreducibility follows from

$$\text{Disc}_u(\Delta_{A,B}) = 64(B^2 - 3Av)^3,$$

since $B^2 - 3Av$ is not a square in $\mathbb{C}(c, v)$. Bézout coefficients for A, B combine the equations

$$3At^2 + 2Bt + v = 0, \quad Bt^2 + 2vt - 6u = 0$$

into a monic quadratic equation for t , proving finiteness. The identities

$$B^2 - 3Av = H^2, \quad -18Au - Bv = 2tH^2$$

give birationality and show that $H^2\mathbb{C}[c, t]$ lies in the conductor.

At the generic point of $H = 0$, use H as transverse coordinate. Over $K = \mathbb{C}(c)$, the inclusion is

$$K[H^2, H^3] \subset K[H],$$

whose conductor is $H^2K[H]$. Every global conductor element therefore has H -valuation at least two. This proves equality and the stated contraction. $\square$

Thus the reduced conductor support

$$L_{A,B} = V(3A(c)t + B(c))$$

is intrinsic to the discriminant surface, not merely detected by a chosen gradient calculation.

F.2. Marked arrangements. Assume A is squarefree and a nonempty subset of its roots marks planes on which an infinity sheet is missing. In the normalization these planes pull back to vertical lines $M_\rho = (c = \rho)$.

Theorem F.2 (Stable conductor-arrangement classification). *Two marked arrangements*

$$\left(\mathbb{A}^2, L_{A,B}, \{M_\rho\}_{\rho \in S}\right) \quad \text{and} \quad \left(\mathbb{A}^2, L_{A',B'}, \{M_{\rho'}\}_{\rho' \in S'}\right)$$

become isomorphic after multiplication by an affine space if and only if there exist

$$\gamma(c) = \lambda c + \mu, \quad \lambda \neq 0, \quad \nu, \kappa \in \mathbb{C}^*, \quad h(c) \in \mathbb{C}[c],$$

such that γ carries the marked roots to the marked roots and

$$A'(\gamma(c))\nu = \kappa A(c),$$

$$B'(\gamma(c)) + 3A'(\gamma(c))h(c) = \kappa B(c).$$

Stabilization introduces no additional transformations.

Proof. If a cylinder automorphism has first coordinate C , mapping one marked line to another gives equality of principal prime ideals and hence

$$C - \rho' = \lambda(c - \rho).$$

Thus $C = \lambda c + \mu$, independent of t and of all stabilization variables. The conductor curves are irreducible principal divisors, so

$$3A'(C)T + B'(C) = \kappa(3A(c)t + B(c)).$$

It follows that $T = \nu(c)t + h(c)$. Factoring the Jacobian determinant of the cylinder automorphism shows that $\nu(c)$ is a unit, hence constant. Coefficient comparison gives the two equations. Conversely, they define the required triangular automorphism without using any stabilization variable. $\square$

For squarefree A , the complete data are therefore the affine configuration of its roots, with represented/missing markings, and the projective vector

$$[B(\rho)]_{\rho \in Z(A)}$$

modulo the finite permutations induced by affine symmetries. Equivalently, after compactifying the t -direction, the boundary tangent at a root ρ is

$$\tau_\rho = -\frac{3A'(\rho)}{B(\rho)},$$

and the projective vector of these slopes is intrinsic.

F.3. Consequences for the q -family. For

$$A(c) = c(c+1), \quad B_q(c) = qc^2 - 4c - 2,$$

the roots 0 and -1 are respectively represented and missing, and

$$B_q(0) = -2, \quad B_q(-1) = q + 2.$$

Thus, for $q \neq -2$,

$$q = 2 \frac{\tau_0}{\tau_{-1}} - 2.$$

At $q = -2$, the discriminant gains a plane factor and the residual component is singular at $(u, v, c) = (0, 0, -1)$; this separates the exceptional member intrinsically. These calculations give another proof of the complete stable q -classification in the body.

F.4. A three-dimensional degree-fifteen family. Let

$$s \in \mathbb{C} \setminus \{0, 1\}, \quad d_1, d_2 \in \mathbb{C}^*,$$

and set

$$A_s(c) = \frac{c(c+1)(c+s)}{s}.$$

There is a unique cubic B_{s,d_1,d_2} satisfying

$$B(0) = -2, \quad B'(0) = -2A_s''(0), \quad B(-1) = d_1, \quad B(-s) = d_2.$$

Explicitly,

$$B = -2 - \frac{4(s+1)}{s}c + b_2c^2 + b_3c^3,$$

where

$$b_2 = \frac{d_1s^3 - d_2 - 2s^3 - 4s^2 + 4s + 2}{s^2(s-1)}, \quad b_3 = \frac{d_1s^2 - d_2 - 2s^2 + 2}{s^2(s-1)}.$$

Theorem F.3 (Degree-fifteen stable moduli). *The associated maps are polynomial Keller maps of component degrees $(15, 14, 4)$, generic degree three, and generic monodromy S_3 . Their stable equivalence relation is exactly*

$$(s, d_1, d_2) \sim (1/s, d_2, d_1).$$

Consequently degree fifteen contains a three-dimensional stable-moduli family, modulo this finite involution.

Proof. The jet conditions at $c = 0$ are precisely the pole-cancellation conditions in the cubic frame. The represented root is 0, while the two missing roots are $-1, -s$. By theorem F.2, an equivalence must fix the represented root and either fix or interchange the missing roots. This gives the identity or the displayed involution. The involution is realized by

$$(x, y, z) \mapsto (x/s, sy, s^2 z)$$

and

$$(a, b, c) \mapsto (s^2 a, sb, c/s).$$

□

Theorem F.4 (Growing stable moduli). *For every $n \geq 2$, the cubic frame contains a $(2n-3)$ -dimensional family of pairwise stably inequivalent generic-degree-three Keller maps of ordinary degree $4n + 3$.*

Proof. Take squarefree A of degree n , with one represented root and $n - 1$ missing roots. Normalize the represented root to 0, impose

$$A'(0) = 1, \quad B(0) = -2, \quad B'(0) = -2A''(0),$$

and prescribe arbitrary nonzero values of B at the other roots. Hermite interpolation produces B of degree at most n . The affine root configuration contributes $n - 2$ parameters, and the projective conductor-value vector contributes $n - 1$. The classification theorem identifies only a finite permutation quotient, giving $(n - 2) + (n - 1) = 2n - 3$ moduli. The leading terms have ordinary degree $4n + 3$. □

Theorem F.5 (Degree threshold inside the cubic frame). *Every polynomial member of ordinary degree at most ten in the full cubic frame is polynomially source-target equivalent to the degree-seven base map. Degree eleven is the first degree at which genuine moduli occur in this frame.*

Proof. If $m = \deg A$ and $n = \deg B$, their unique leading contributions have degrees

$$(4m + 3, 4m + 2) \quad \text{and} \quad (4n + 2, 4n + 1)$$

in the first two coordinates. Degree at most ten forces

$$\deg A = 1, \quad \deg B \leq 2.$$

After centering and scaling,

$$A(C) = C, \quad B(C) = -2 + b_1 C + b_2 C^2.$$

The coefficient b_1 is removed by the source translation responsible for the w_{20} -gauge. The b_2 -family is removed by the explicit locally nilpotent flow already described in the classification of the $\alpha = 0$ line. The q -family supplies inequivalent degree-eleven members. $\square$

F.5. Formal arcs and global separation. For fixed q , take

$$A_s(c) = c + sc^2, \quad B_{s,q}(c) = -2 - 4sc + qs^2c^2.$$

The special fiber $s = 0$ is the base map, and every $s \neq 0$ fiber lies in the stable orbit indexed by q . The coefficientwise formal source trivialization has, through order three, initial degrees

$$(19, 17, 19), \quad (37, 35, 37), \quad (55, 53, 55).$$

The first coefficient is independent of q , and q first occurs in the second.

Proposition F.6 (Formal-local versus stable separation). *All q -arcs have the same first-order formal tangent and are source-trivial through the three computed formal orders, while their punctured fibers remain pairwise stably inequivalent. Thus the stable conductor modulus is invisible to this local formal data.*

Proof. The formal inverse recursion gives the displayed coefficients directly. Stable inequivalence of the punctured fibers follows from the intrinsic ratio of conductor values. No assertion is made here about a minimal degree formula for all higher formal orders. $\square$

APPENDIX G. THE BOUNDARY-FRAMED LOGARITHMIC DEFORMATION COMPLEX

At the degenerate reciprocal member $g = 1$, consider the finite flat cubic completion

$$X_0 = V(2pu^3 - qu^2v + 2uv^2 - rv^3) \subset \mathbb{A}_{p,q,r}^3 \times \mathbb{P}_{[u:v]}^1.$$

On $u \neq 0$, put $t = v/u$ and

$$J = \frac{3rt^2 - 4t + q}{2}.$$

Then (t, r, J) are coordinates on X_0 , with

$$q = 2J + 4t - 3rt^2, \quad p = tJ + t^2 - rt^3.$$

The repeated-root divisor is $D_0 = (J = 0)$, and the triple-root curve is

$$C_0 = (J = 0, 2 - 3rt = 0).$$

Proposition G.1 (Exact logarithmic complex). *For deformations of*

$$\pi_0 : (X_0, D_0) \longrightarrow \mathbb{A}^3$$

with target fixed and D_0 marked, the nontrivial part of the logarithmic tangent differential is

$$N = \begin{pmatrix} J & 0 \\ 2 - 3rt & J \end{pmatrix}.$$

It is injective, and

$$H^1(\mathbb{T}_{\pi_0}^{\log}) = \text{coker } N, \quad H^i(\mathbb{T}_{\pi_0}^{\log}) = 0 \quad (i \neq 1).$$

Moreover,

$$\text{Fitt}_0(\text{coker } N) = (J^2), \quad \text{Fitt}_1(\text{coker } N) = (J, 2 - 3rt).$$

Proof. A basis for $T_{X_0}(-\log D_0)$ is

$$\partial_t, \quad \partial_r, \quad J\partial_J.$$

The ∂_r -to- ∂_r summand splits. In the remaining target coordinates

$$U = \delta p - \frac{t}{2}\delta q, \quad V = \frac{1}{2}\delta q,$$

direct differentiation gives N . Since $\det N = J^2$ in the domain $\mathbb{C}[t, r, J]$, N is injective. Affineness gives the stated hypercohomology, and the Fitting ideals are the minors of the displayed matrix. $\square$

For

$$g_\epsilon(p) = 1 + \epsilon p^2 \eta(p), \quad \epsilon^2 = 0,$$

the induced class is

$$[-p^3 \eta(p)]e_U \in \text{coker } N.$$

It is nonzero for $\eta \neq 0$: restriction to $J = 0$ sends

$$p \mapsto t^2(1 - rt)$$

and embeds $\mathbb{C}[p]$ into $\mathbb{C}[t, r]$. These directions integrate to the exact reciprocal family.

Remark G.2 (Why the framing matters). If arbitrary target vector fields are restored, the target field $-p^3 \eta(p) \partial_p$ kills the displayed infinitesimal class. The calculation therefore describes the fixed-target, boundary-marked deformation problem, or any finite-type moduli problem imposing equivalent framing. It does not claim that the same tangent vector survives unrestricted affine target gauge.

AI AND COMPUTATIONAL DISCLOSURE

AI systems were used extensively in exploration, proof drafting, symbolic program development, source organization, and manuscript editing. They are not authors. The mathematical evidence for the computer-checked identities is the exact verifier and reproducible output described in section 7, not the output of a language model. Human specialist review has not yet been completed.

REFERENCES

1. Levent Alpöge, *Post announcing an explicit counterexample to the jacobian conjecture*, X post.
2. Adrien Dubouloz and Karol Palka, *The jacobian conjecture fails for pseudo-planes*, *Advances in Mathematics* **339** (2018), 248–284.
3. Alexis Gallagher, *The jacobian counterexample, explained*, The page credits substantial mathematical, expository, and computational assistance to GPT-5.6-sol.
4. Zbigniew Jelonek, *The set of points at which a polynomial map is not proper*, *Annales Polonici Mathematici* **58** (1993), no. 3, 259–266.
5. Nathaniel Monson, *Cubic marked-root covers: Normal forms, equivalence, and stable uniqueness*, Companion manuscript.
6. Terence Tao, *A digestion of the jacobian conjecture counterexample*.
7. Ulam AI, *A counterexample to the jacobian conjecture*.